Indirect reciprocity with inertial reputation updating

Introduction, Storyline 1: from instantaneous to cumulative reputational change

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,8,18]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[12,19,20]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[18,21,22]. A remedy must therefore keep justified punishers in good standing without letting genuine defectors escape condemnation.

Part of this fragility lies in how reputations respond to evidence. In the standard framework, each new assessment replaces an individual's image, so reputations change instantaneously and are determined entirely by the latest observation[8,12]. A single disputed act of punishment is then enough to overturn a standing built over many interactions. Models that draw on more evidence before assigning a reputation, through quantitative scores, simple records, graded reputations or the aggregation of several observations, can make cooperation more robust[21,23–27]. Most of them, however, treat reputational change as a single symmetric process. Positive and negative evidence enter through the same score, threshold or observation window, and evidence affects treatment only once a threshold is crossed. Human evaluation departs from both features. Impressions of others are revised asymmetrically, although the direction of the asymmetry depends on context[28–31]. Institutions, too, grade their response and treat the two directions differently, escalating sanctions only with repeated violations[32,33] and restoring an offender's standing only after years without a further offence[34]. Two questions therefore remain open. Should losing a good reputation require as much evidence as regaining one? And how should individuals be treated while evidence about them is still accumulating? Both bear directly on the punishment dilemma, because buffering negative evidence protects justified punishers but may also shelter those who defect deliberately, provided they do so only occasionally[21,35].

Here, we develop a model of indirect reciprocity in which reputations change through accumulated evidence rather than instantaneously (Fig. 1). Negative assessments move an individual in good standing through a sequence of warning states before the reputation turns bad, and positive assessments move an individual in bad standing through recovery states before it turns good. The two paths can differ in

length, so that losing and regaining standing may require different amounts of evidence. Each intermediate state is interpreted as good or bad with a probability that sets how strongly incomplete evidence already affects treatment. The model thus treats two properties of reputational change as separate design choices, namely how much evidence is needed to change a reputation in each direction and how incomplete evidence is expressed in the meantime.

We find that the allocation of inertia between the two directions, rather than its total amount, determines whether private observers agree with one another and still distinguish cooperators from defectors. The beneficial direction depends on the norm. Compact architectures, selected solely by these reputational criteria, make cooperation stable and easy to reach under each of four classical norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all. We further show that the tolerance that protects justified punishers can also shelter opportunists who defect only occasionally, a vulnerability that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Reputation-based cooperation therefore depends not only on which behaviours a society judges as good, but also on how readily it revises those judgements in each direction and on how it treats individuals before a judgement changes.

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